Simranjeet Singh

Bangalore, India
www.simranchawla.com | github.com/simranjeetc | www.linkedin.com/in/simranjeetchawla

Platform Architect | Distributed Systems | Cloud & AI Platforms

Engineer with 17+ years building large-scale distributed systems and cloud platforms, operating at principal-level scope. Leads high-ambiguity, cross-functional initiatives spanning architecture, reliability, and developer platform foundations.

HIGHLIGHTS

- Architected and drove org-wide adoption of platform libraries — circuit breaker, metrics, resilient clients, HIPAA-compliant auditing — now used across multiple product teams.
- Led decomposition of a 15-domain modular monolith into 6 microservices across 4 delivery phases.
- Separated Kafka consumer and REST API workloads across 9 services to eliminate GC contention and enable independent scaling.
- Designed fair-queueing and background-processing architectures for high-throughput multi-tenant workloads.
- Contributed to an AI orchestration + RAG pipeline spanning chat and voice channels.
- Stabilized JVM workloads at 40GB+ heaps in production and established cluster-wide runtime standards.
- Architected a 5TB cross-account data migration pipeline using AWS Glue, Spark, and Apache Airflow.

AI-FIRST ENGINEERING

- Delivery built on agentic tooling — Claude Code, GitHub Copilot, OpenCode, and CLI agents — rather than occasional AI use.
- Spec-driven development with BMAD and OpenSpec across the full delivery cycle, from requirements through implementation.

EXPERIENCE

JPMorgan Chase - Lead Software Engineer (VP)

August 2026 - September 2026

- Architected and deployed 5TB data migration pipeline across AWS accounts using Glue, Spark, and Apache Airflow

Nextiva - Staff Engineer

March 2025 - July 2026

- Drove separation of Kafka consumer workloads from REST API deployments across 9 core services — including a single service running 5 concurrent consumer tasks within a pod configured up to 60GB memory — eliminating shared GC pressure and thread pool contention, enabling independent scaling of event-processing and API tiers, and reducing blast radius of consumer failures on user-facing traffic.
- Architected a phased decomposition of a 15-domain modular monolith (memory limits up to 28GB per pod) into 6 discrete microservices across 4 delivery phases, defining extraction sequencing, dual-write migration strategy, and shared library boundaries to reduce production load concentration and establish clear team ownership boundaries.
- Architected reusable platform libraries (circuit breaker, metrics, resilient clients, HIPAA-compliant auditing, telemetry) and drove their adoption across multiple product teams.
- Drove the design and implementation of an AI orchestration layer and RAG pipeline — covering chunking, embedding, tool execution, hybrid search, semantic caching, and LLM synthesis across chat and voice channels.
- Built the retrieval and ranking path of the RAG system in Python — MongoDB Atlas vector search, Redis caching, and result ranking and merging — running in production.

OpenText - Software Architect

December 2021 - February 2025

- Designed fair scheduling for multi-tenant execution to prevent starvation and improve tenant fairness.
- Scaled data processing to 30TB/day by architecting background-processing for large query workloads.

- Led the SDM-to-FAS cloud migration, coordinating the FAS and SDM engineering teams and their stakeholders to align scope and delivery.
- Introduced distributed tracing primitives (correlation IDs) to improve end-to-end diagnostics.

OpenText - Technical Lead June 2019 - November 2021

- Improved distributed job-scheduler behavior for cancellation/delete-heavy workloads.
- Modernized Kubernetes ingress stack for better platform consistency.
- Introduced Testcontainers-based integration testing to improve confidence and maintainability.

OpenText - Technical Lead January 2019 - June 2019

- Built a custom Kafka connector for Vertica to RDBMS synchronization to enable near real-time data flow and improved read performance.
- Developed backend framework capabilities for multi-dashlet support.
- Supported release war-room performance triage and rapid issue closure.

Earlier Roles

- Software Specialist, OpenText (June 2016 - January 2019)
- Member Technical Staff, Ness Technologies (March 2012 - June 2016)
- Product Development Engineer, Symphony (November 2011 - March 2012)
- Project Engineer, Wipro Technologies (December 2008 - November 2011)

TECHNICAL SKILLS

Languages: Java, Groovy(Grails), TypeScript, Bash, Python, Go, SQL

AI/ML: GPT-4o, Gemini, LangChain, LangGraph, RAG (hybrid search + RRF)

AI/ML Infra: MongoDB Vector Search, Redis Vector Search, STT/TTS (Deepgram, Google Cloud TTS, NVIDIA NeMo)

AI-Assisted Development: Claude Code, GitHub Copilot, OpenCode, CLI agents

Spec-Driven Development: BMAD, OpenSpec

Frameworks: Spring, Hibernate, Angular, Flask

Cloud/DevOps: GCP, AWS, S3, Glue, Docker, Kubernetes, Jenkins, CI/CD

Monitoring & Logging: Prometheus, Grafana, ELK Stack, Datadog

Data: BigQuery, PostgreSQL, MongoDB, Redis, Elasticsearch, Oracle, MySQL, MS-SQL, Vertica, Spark, Airflow

Messaging/Integration: Kafka, RabbitMQ, REST, gRPC

Security/Identity: OAuth2, SAML, JWT, SSL/TLS, OWASP practices

Engineering Tools: Git, GitHub, GitLab, JIRA, JUnit, Mockito, SonarQube

EDUCATION

B.Tech, Sastra University (2004 - 2008)